

Mike Mabey, Ph.D.

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EXPERIENCE

Backend Software Engineer

Macabacus Inc.

Apr 2025 – Present
(remote) Salt Lake City, UT

- Converted code that used **Celery** to use **Taskiq**, allowing for asynchronous queuing and processing of long-running tasks, improving the responsiveness of the web interface, and decreasing API server's load.
- Helped maintain and improve the internal, backend API, comprised of **FastAPI** and **MySQL** running on **Azure**.

Senior Software Engineer, Back End

Capital One

Jan 2024 – Mar 2025
New York City, NY / (remote) Salt Lake City, UT

- Identified a gap in API developers' workflow that was manual, error prone, and time consuming, then created a tool to improve schema changes to be programmatic and guaranteed to be consistent with code.
- Refactored a **streamz** data pipeline to support more than one customer. Afterward, onboarded a client with more than five times any previous traffic volume in an effort projected to save \$2+ million annually.
- Discovered a bottleneck that could not scale horizontally, then purposefully applied Python's **asyncio** library and **SQS**'s batched submission to increase max throughput from 335 to >7,000 transactions per second.
- Created many APIs on the company's data exchange that both met our customers' needs and fit within our data pipeline's capabilities.
- Assisted my teammates with their Python knowledge to improve their work, such as demonstrating how to use **Jinja** templates with **Pydantic** models to simplify the creation of dynamic text output.

Backend Engineer 3 - Product

Cedar Cares, Inc.

Jul 2019 – Jul 2023
Salt Lake City, UT

- Technical point-of-contact while integrating with two major electronic health record (EHR) systems.
- Eliminated the need for teammate engineers to manually validate composed JSON payloads by conceiving, authoring, and open-sourcing a **Python** library for generating spec-compliant payloads (🐍 pyredox).
- Reduced client data refresh times from 30 minutes (average) down to a few seconds using **Domain-Driven Development** (DDD) patterns to refactor a batch-based **Python** service to be event-based.
- Implemented insurance claim adjudication tools with a stack of **Python**, **gRPC/protobufs**, **AWS (EC2, SQS, S3, and KMS)**, **MariaDB/Postgres**, **OpenCV**, **OAuth**, as well as the **X12 EDI** and **FHIR** data formats.
- Wrote a customer-facing API using **FastAPI** with accompanying documentation in **Swagger**.

Computer Scientist (U.S. Army Civilian)

Data Science Directorate, Network Enterprise Technology Command (NETCOM)

Dec 2017 – Jun 2019
Phoenix, AZ

- Implemented a dashboard that computed and visualized, for the first time ever, the command's global performance on servicing help desk tickets using **Python**, **Flask**, **Vue.js**, **pandas**, **Plotly**, **Celery**, and **redis**.
- Shortened the development cycle for big data analytic apps by automating the build and deployment process using **Python** and GitLab's **Continuous Integration** utility.
- Saved the command \$3 million annually as technical liaison to Sandia National Laboratories by recommending to discontinue their contract because it had proved unfruitful for multiple consecutive years.
- Initiated a culture in the Directorate of using **git**, **GitLab**, and DevOps methods and established internal best practices for collaborating on code development and documenting lessons learned.

EDUCATION

Ph.D. Computer Science — Information Assurance

Arizona State University

Dec 2017

Tempe, AZ

M.S. Computer Science — Information Assurance

Arizona State University

Aug 2011

Tempe, AZ

B.S. Computer Science — Information Systems

Utah State University

May 2009

Logan, UT